

Chapter Highlights

- 1 In molecular biology, smaller models are outperforming larger ones.** [MSAPairformer](#), a 111-million-parameter protein language model, outperformed previous leading methods on the benchmark, ProteinGym; and [GPN-Star](#), a 200-million-parameter genomics model, outperformed a model with 40 billion parameters.
- 2 Virtual cell models emerged as a new frontier in 2025, with major releases including Evo 2 from the Arc Institute, STATE, and DeepMind's AlphaGenome.** These models aim to predict cellular responses to drugs and genetic perturbations without running wet-lab experiments, though current systems still require experimental validation.
- 3 Like other areas of AI, biological model development is increasingly bottlenecked on data rather than architecture.** With cofolding models now representing all structure types in the Protein Data Bank, 2025 saw a turn toward distilled datasets of AI-predicted structures and training on combined experimental data sources, expanding training sets from hundreds of thousands of entries to tens of millions.
- 4 AI tools that automatically generate clinical notes from patient visits saw broad adoption in 2025.** Across multiple hospital systems, physicians reported they were spending up to 83% less time writing notes, experiencing significant reductions in burnout, with one hospital system reporting a 112% return on investment.
- 5 The FDA authorized 258 AI medical devices in 2025, most through pathways that do not require new clinical trials.** The vast majority entered the market via device-modification pathways that rely on existing safety and efficacy evidence rather than new randomized trials, with only 2.4% of devices with clinical studies supported by randomized trial data.
- 6 A multi-agent AI system scored 85.5% on complex published case studies, versus 20% for unaided physicians.** Microsoft's AI Diagnostic Orchestrator, paired with OpenAI's o3, was tested on challenging cases drawn from the medical literature against physicians working without their usual tools. Multi-agent frameworks more broadly have shown diagnostic accuracy gains of 7% to over 60% over single-agent baselines.
- 7 AI-generated summaries now appear at the top of 84% to 92% of health-related Google searches.** Symptom and common health questions trigger an AI Overview 92% of the time, followed by treatment and condition queries. These summaries are now a routine feature of health information searches, shaping the initial interpretation of users' questions.
- 8 Ethics discussion in medical AI publications more than doubled in 2025, but the conversation is narrow.** Governance dominates the discourse, while algorithm accountability, biosecurity, and global health equity remain underexplored.
- 9 Research interest in medical digital twins is growing fast, and where rigorous trials exist, early results are promising.** In a randomized trial of 150 diabetes patients, 71% achieved healthy blood sugar levels over one year while safely reducing their medications.

6.1 The Central Dogma

AI models for molecular biology span the pathway from gene sequence to protein structure to therapeutic design. This section tracks advances in protein language models, structure prediction, protein design, virtual cell models, and multimodal foundation models for biomedical discovery. The analysis draws on PubMed publication counts, benchmark evaluations including [ProteinGym](#) and [FoldBench](#), and model release data from 2024 and 2025. A recurring pattern across these areas is the tension between scale and specialization. In several areas, smaller or more targeted models matched or outperformed larger general-purpose systems.

Research Volume

AI-driven protein research grew approximately 71% between 2024 and 2025 (Figure 6.1.1). Total publications across four categories—function prediction, protein structure prediction, protein-drug interactions, and synthetic protein design—rose from 2,259 in 2024 to 3,855 in 2025. Protein-drug interactions represented the largest share of output in both years, accounting for 49.9% of papers in 2024 and rising to 54.4% in 2025. Protein structure prediction constituted the second-largest category in 2024 at 28.7%, though its relative share declined to 23.9% in 2025. Function prediction and synthetic protein design each remained comparatively stable, with function prediction increasing from 9.7% to 10.4% and synthetic protein design decreasing slightly from 11.7% to 11.3%. The shift in relative share toward protein-drug interactions, even as absolute counts grew across all categories, may reflect maturing structure prediction methods and growing interest in therapeutics applications. Publications focused specifically on AI for drug discovery have followed a similar upward trajectory (Figure 6.1.2).

Number of AI-driven protein research publications, 2024 vs. 2025

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

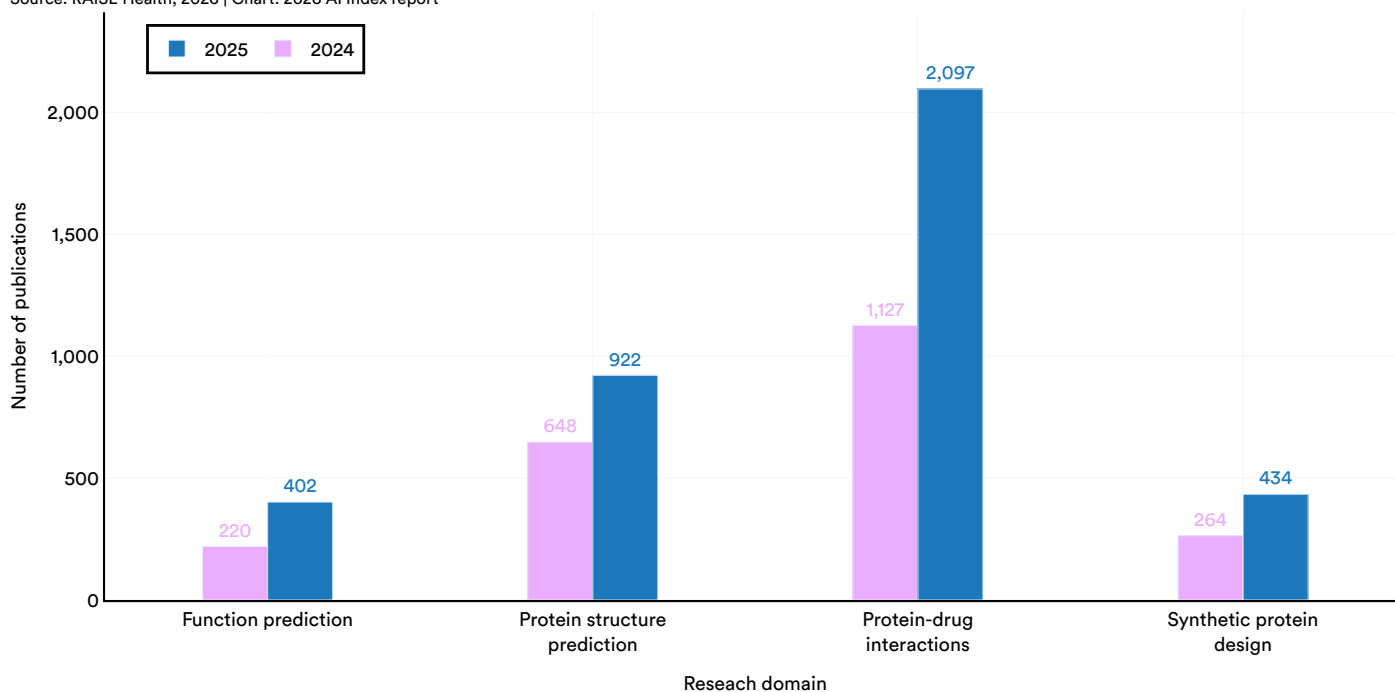


Figure 6.1.1

Number of publications on AI for drug discovery, 2018–25

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

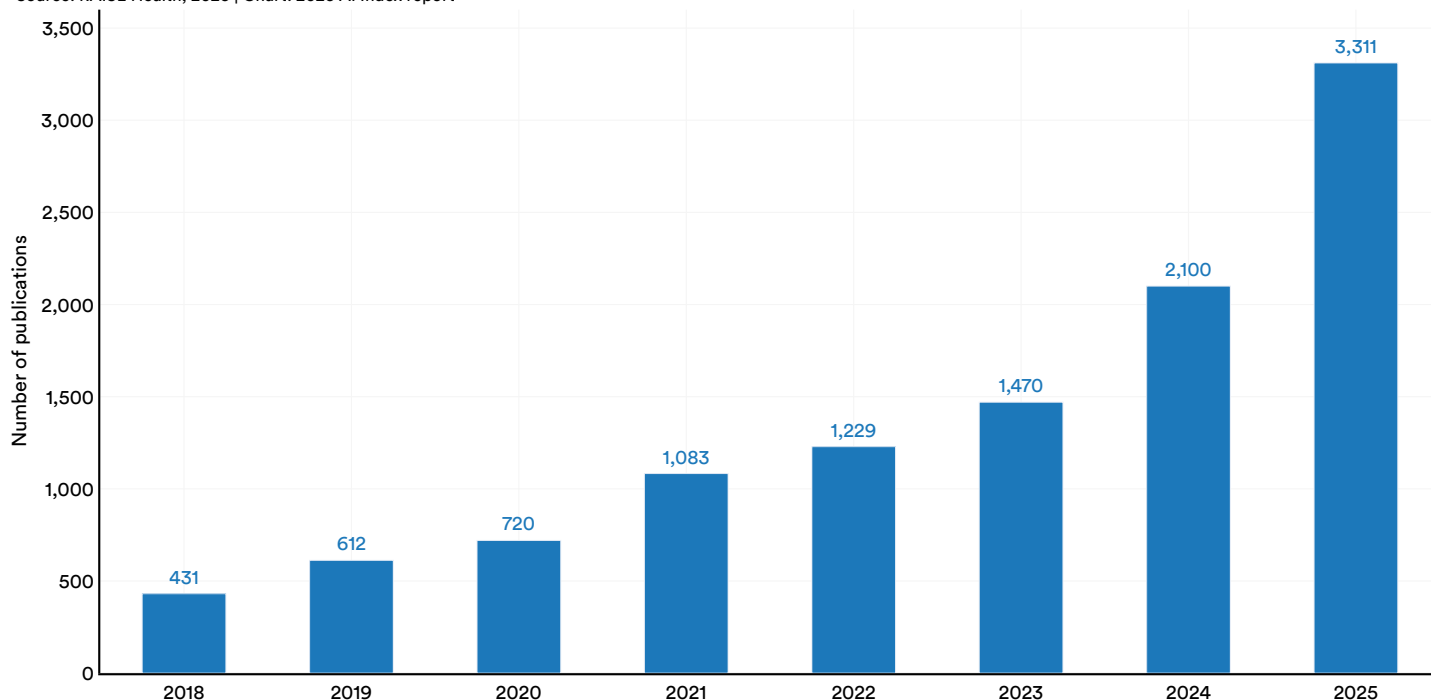


Figure 6.1.2

Public Datasets

Molecular and Cellular Biology

Demand for training data has continued to grow as AI models have gained further adoption in biology. Rapidly collecting new biological data is typically time-consuming and expensive. In 2025, biological AI models increasingly trained on multiple datasets with different types of experimental measurements. Several cofolding methods (where two or more molecules are modeled simultaneously), for example, began training on both structural data from the [Protein Data Bank](#) (PDB) and experimental small-molecule binding affinity measurements from repositories such as [PubChem](#), [ChEMBL](#), and [BindingDB](#).

The scale of publicly available biological datasets has grown by several orders of magnitude since the PDB's founding in 1971, though direct size comparisons across databases should be interpreted with caution. The unit of measurement differs by source: An “entry” may represent an experimentally solved protein structure, a bioactivity measurement, a gene sequence, or a single-cell observation.

Other models are trained on synthetically generated data. [AlphaFold 3](#) and its open-source replications, including [Boltz-2](#) and [OpenFold3](#), all train on “self-distillation” datasets of predicted protein structures from [AlphaFold 2](#) that have been filtered for quality. Meta FAIR released [Open Molecules 2025](#) (OMol25), a dataset of over 100 million quantum mechanics calculations of molecules.

New experimental datasets also debuted in 2025. [Tahoe-100M](#), the largest publicly released, single-cell sequencing dataset, contains measurements from over 50 cancer cell types exposed to more than 1,100 drugs. [BaseData](#) features over 9.8 billion genes obtained through metagenomic mining (Figure 6.1.3).

Size of public datasets for molecular and cellular biology

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

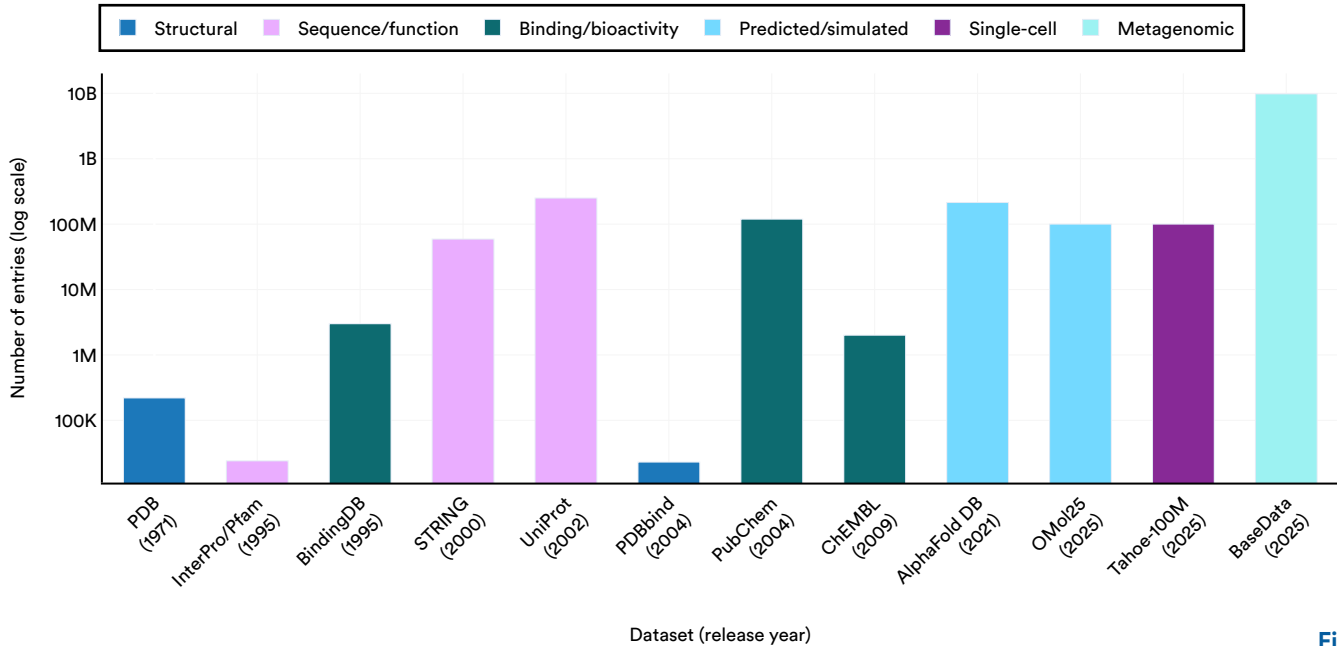


Figure 6.1.3

Data for Biomedical Vision-Language Models

Training biomedical vision-language models requires large repositories of images and captions that can be transformed into a continual pretraining dataset. In the general domain, data scaling is often considered a mature or saturated research direction, but this does not appear to hold in the biomedical setting. Newer datasets extend beyond a single specialty and incorporate a broader range of modalities and biomedical domains (Figure 6.1.4).

Size of select biomedical datasets at release, 2020–25

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

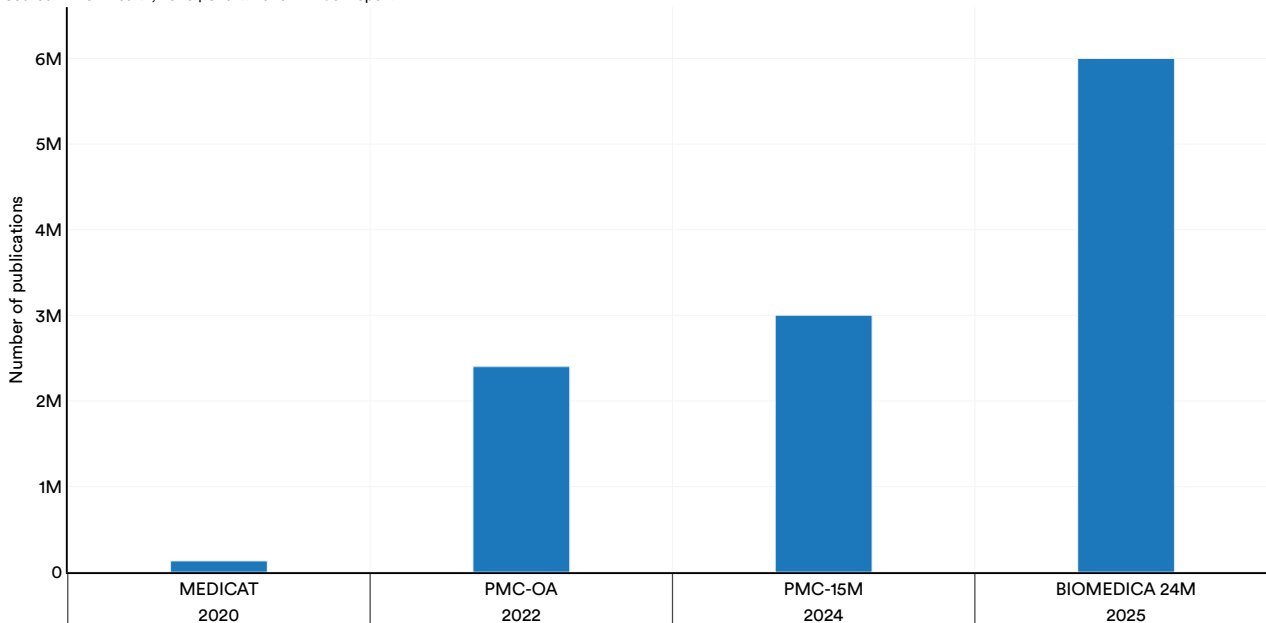


Figure 6.1.4

Sequence-Based Models: Protein Language Models

The trend in protein language models (PLMs) shifted in 2025 from scaling model size to improving model efficiency and specialization. In 2024, efforts culminated in the 98-billion-parameter ESM3. In 2025, the focus turned to smaller architectures trained on curated data or augmented with retrieval methods (Figure 6.1.5).

ProteinGym is a comprehensive benchmark suite for protein fitness prediction and design, comprising over 250 standardized deep mutational scanning assays with millions of mutated sequences and curated clinical datasets with expert-annotated mutation effects. [MSAPairformer](#), a 111 million-parameter model trained on multiple sequence alignments and physical constraints, surpassed previous state-of-the-art methods at a fraction of the training and parameter budget (Figure 6.1.6). The [Profluent E1](#) series similarly set new performance standards by combining a smaller model with a retrieval augmented generation (RAG) approach. While certain tasks still benefit from larger models, others—such as predicting cellular localization—appear to vary more with training method and data than with parameter count.



Size of protein sequencing models, 2020–25

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

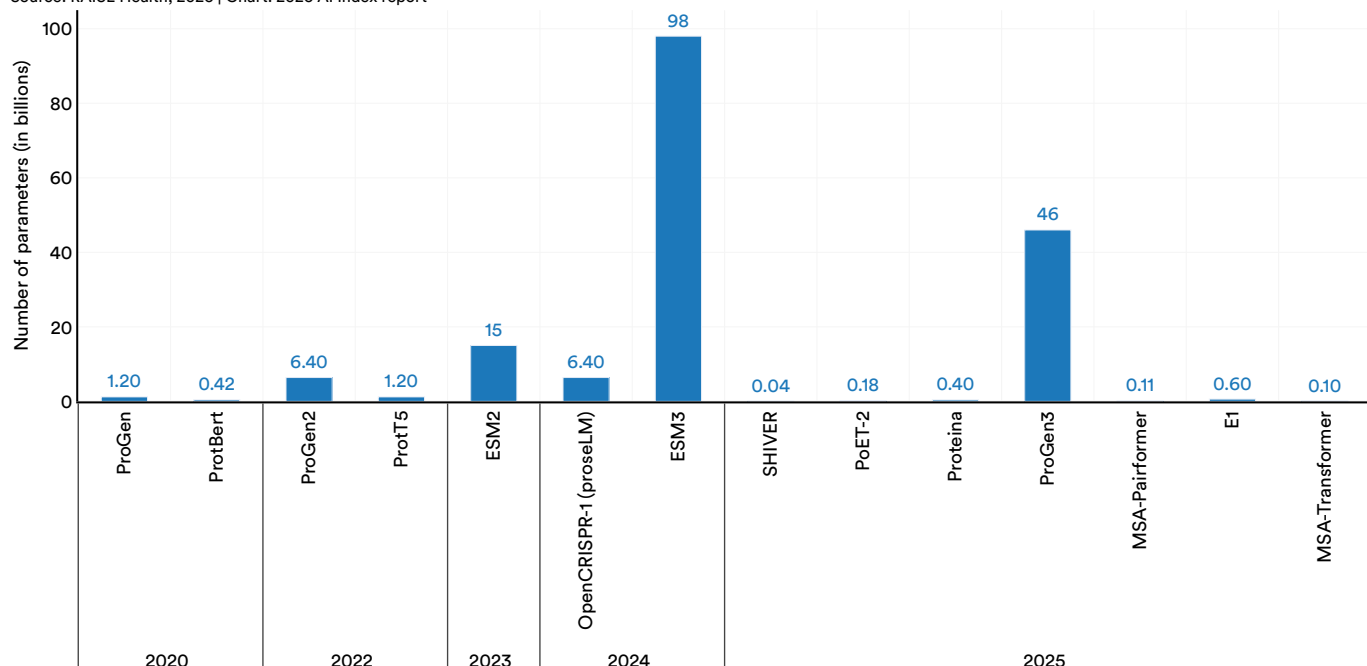


Figure 6.1.5

Performance of protein language models on ProteinGym, 2021–25

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

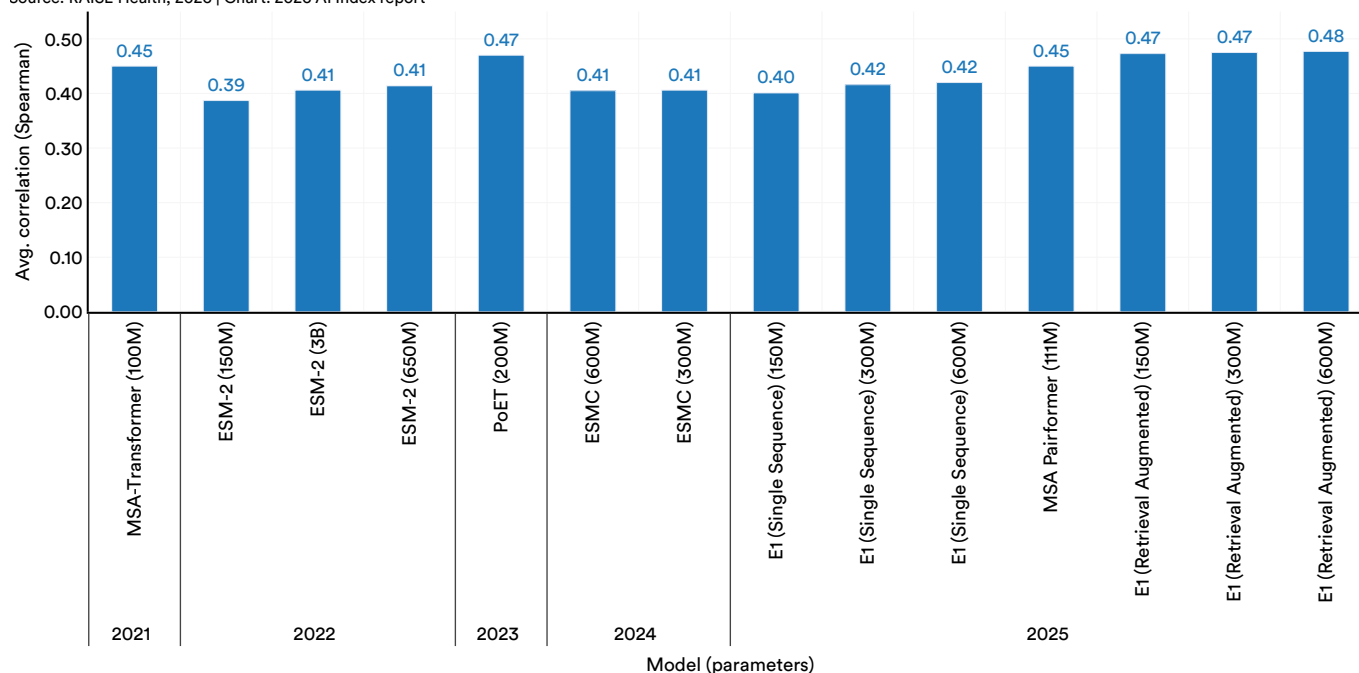


Figure 6.1.6

Beyond benchmark performance, PLMs have also become more task-specific. The [ESM-C series](#) demonstrated that smaller models geared toward a single task, such as representation learning, could be successful without the full feature set of the ESM3 family. A fine-tuned [ProGen model](#) (6 billion parameters) was used to design a novel CRISPR-Cas protein, [OpenCRISPR-1](#), which demonstrated improved specificity relative to standard [SpCas9](#).

Structure Prediction and Cofolding Models

Multiple open-source structure prediction models were released in 2025, inspired by the architecture of AlphaFold 3. These models tackle the task of “cofolding”—predicting the three-dimensional structures formed by combinations of proteins, nucleic acids, drugs, and other biomolecules. While AlphaFold 3 retains a performance advantage on certain tasks, most cofolding models have demonstrated similar performance across protein structure and biomolecular complex modeling tasks. Some, including the Boltz series and OpenFold3, are released under commercially permissive licenses.

Because cofolding models can now represent all structure types available in the PDB, further performance gains will likely require new data sources or deeper extraction of signal from existing data. One strategy is the use of “distilled” datasets, in which AI-predicted protein structures supplement experimentally determined ones, scaling training datasets from hundreds of thousands of entries to tens of millions. Boltz-2 illustrates a complementary approach—training on both structural information from the PDB and experimental binding affinity measurements—to combine specialization and scale (Figure 6.1.7). The four leading cofolding models share a common foundation of experimental and distilled structural data, but they diverge in their use of additional sources such as molecular dynamics simulations, binding affinity measurements, and RNA structure.

Training data sources for cofolding models, 2025

Model	Structural data (experimental)	Structural data (distilled)	Molecular dynamics	Binding affinity	RNA structure
AlphaFold 3	✓	✓		✓	
Boltz-2	✓	✓	✓	✓	
SimpleFold	✓	✓	✓		
OpenFold3	✓	✓		✓	✓

Figure 6.1.6¹

Similar to trends in other areas of AI, bigger models have not necessarily translated to better performance in protein structure prediction. Following the release of AlphaFold 3, subsequent models have converged on a similar parameter scale rather than continuing to grow (Figure 6.1.8). FoldBench is a benchmark that tests whether a model can correctly predict how a small molecule, such as a drug candidate, physically binds to a target protein. AlphaFold 3’s performance on FoldBench has yet to be significantly surpassed even though several larger models have been released since (Figure 6.1.9). These results suggest that data, rather than model size, is an important bottleneck in protein structure prediction.

¹ Rows represent individual cofolding models. Columns indicate whether each model incorporated a given data type during training, including experimentally determined and distilled protein structures, molecular dynamics simulations, binding affinity measurements, and RNA structural data. A check mark indicates that the data type was used.

Size of protein structure prediction models, 2021–25

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

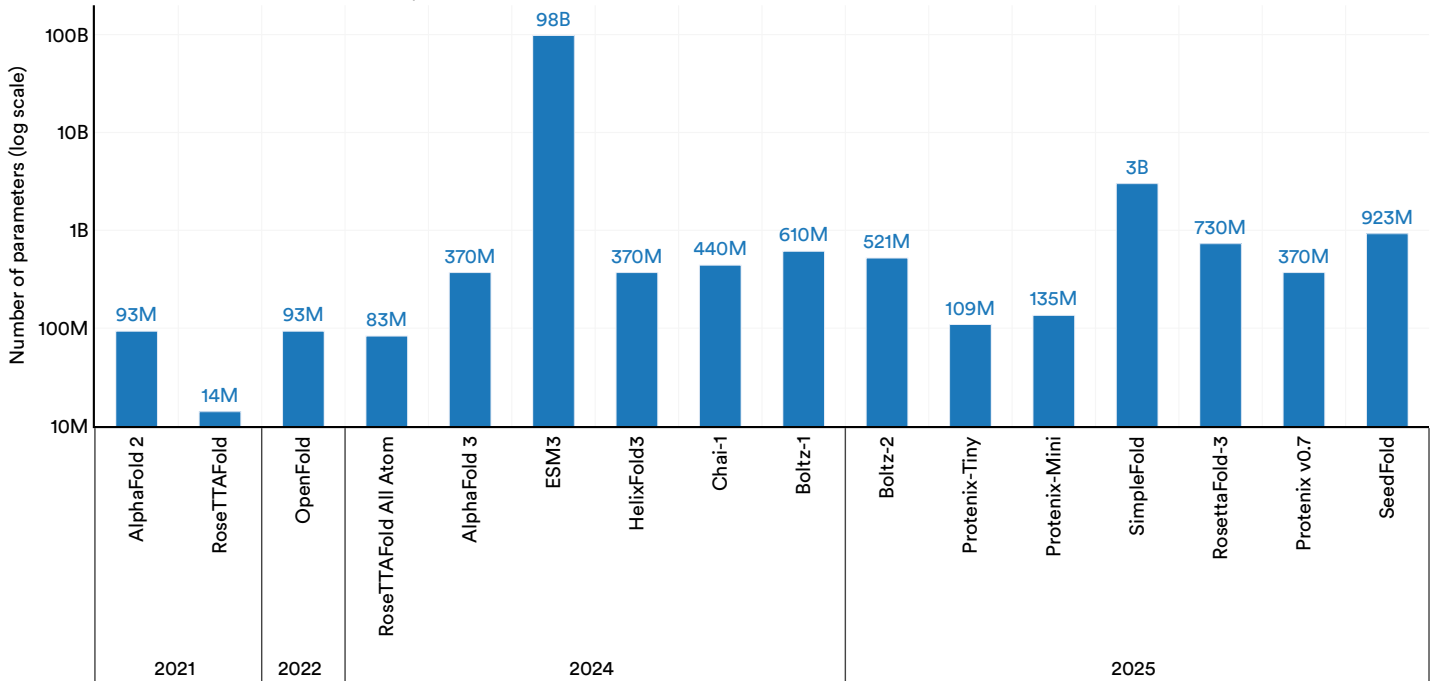


Figure 6.1.8

FoldBench: protein cofolding performance, 2024–25

Source: RAISE Health, 2026 | Chart: 2026 AI Index report

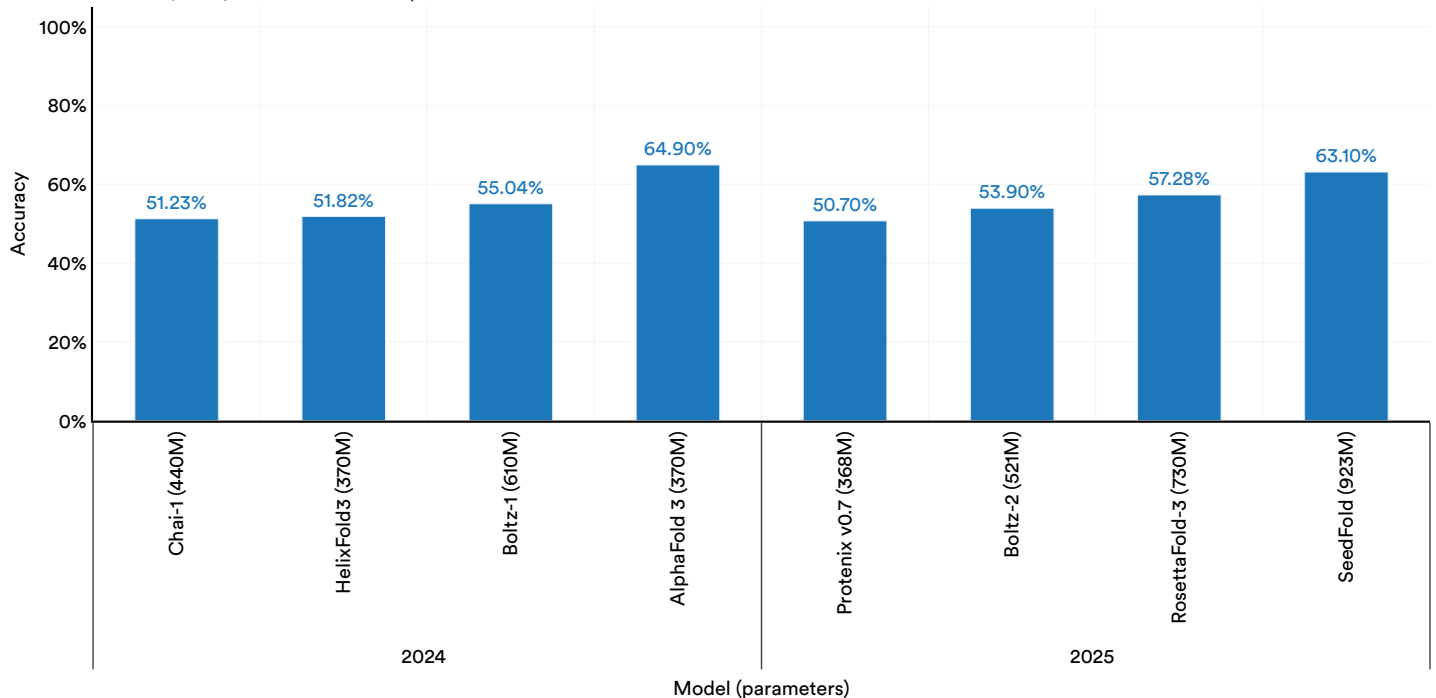


Figure 6.1.9

Protein Design and Generative Models for Therapeutics

Advances in cofolding have enabled a new generation of generative models for protein design, including methods for designing antibodies, nanobodies, and peptides. Approaches range from workflows built around existing structure prediction methods (e.g., BindCraft, Germinal) to models directly trained for generating binders (e.g., RFDiffusion, BoltzGen).

A [protein design challenge hosted by Adatpyv Bio](#) in 2025 provided a controlled comparison. Multiple methods were tested on the task of designing a binder targeting Nipah virus. Of the thousand-plus designs tested, 99 proteins were confirmed to bind, and none neutralized the targeted protein. The specialized method Mosaic, which combines multiple tools with expert tuning, outperformed general-purpose approaches (Figure 6.1.10).

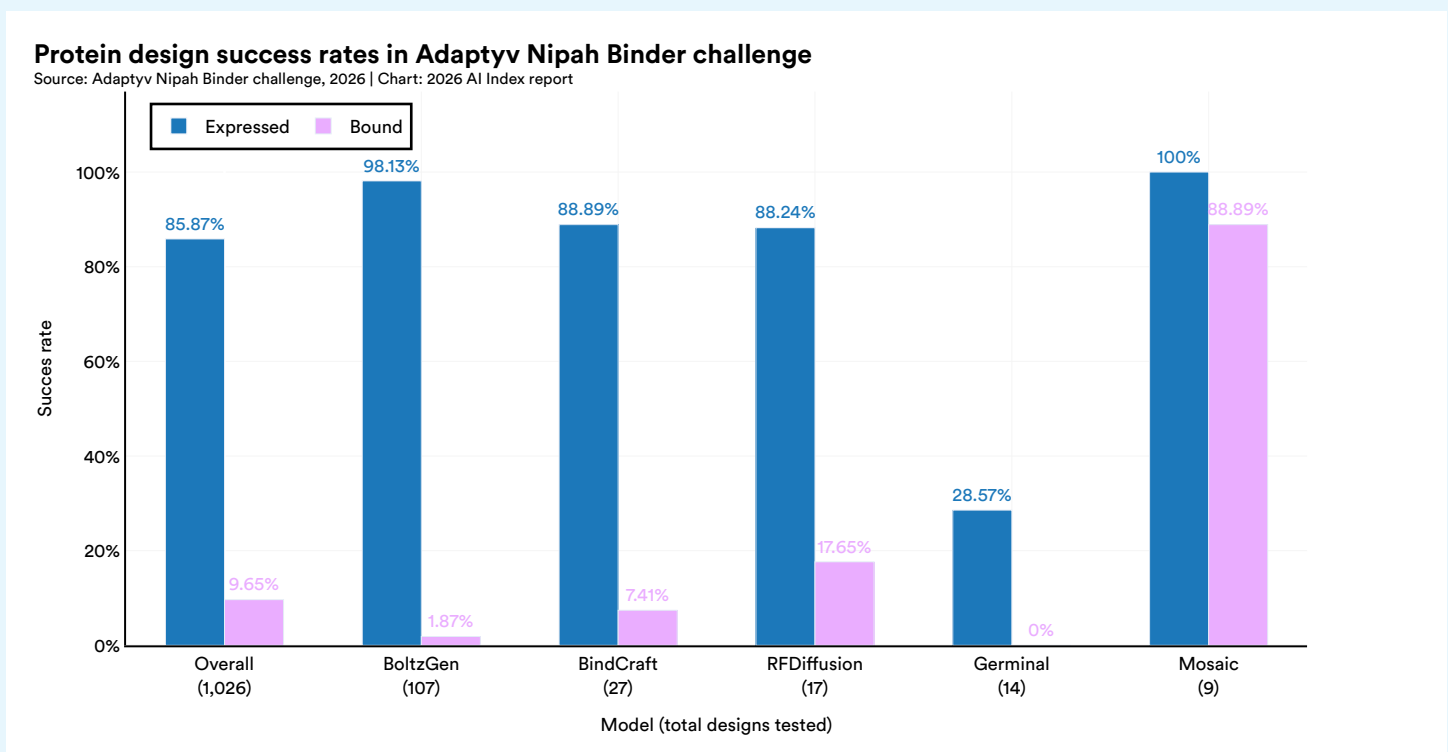


Figure 6.1.10

Virtual Cell Models and Genomic Foundation Models

Research on “virtual cell” models—AI systems that model cellular states and responses to stimuli—increased substantially in 2025, as reflected in growing PubMed publication counts (Figure 6.1.11). Notable releases included [Evo 2](#), a DNA language model from the Arc Institute; [STATE](#), a perturbation-response model; and [AlphaGenome](#), a multimodal model from DeepMind.

However, current virtual cell and genomic foundation models still lag behind smaller, task-specific models on several benchmarks. [GPN-Star](#), a 200-million-parameter model focused on functional and regulatory genomics, [outperformed](#) Evo 2 (40B parameters) on multiple variant effect prediction tasks (Figure 6.1.12). These results suggest that scale alone is not yet sufficient, and that training method and data curation remain important determinants of performance.

HIGHLIGHT:

Automated and Agentic Biomedical Discovery

In 2025, efforts to automate scientific discovery focused on integrating digital reasoning with physical laboratory validation. [Robin](#), an automated discovery framework, linked literature-based hypothesis generation with experimental data analysis, identifying the ROCK inhibitor ripasudil as a novel candidate for dry age-related macular degeneration. [STELLA](#), an autonomous bioinformatics agent, expanded its own technical capabilities by discovering and integrating new software tools rather than relying on manually curated toolsets. [Biomni](#), a general-purpose biomedical AI agent developed at Stanford University, mapped a unified biomedical action space across 25 subfields, integrating 150 specialized tools, 105 software packages, and 59 databases.

Collaborative multiagent frameworks also emerged. Agent Laboratory, developed by AMD and Johns Hopkins, assigns distinct roles to PhD, Postdoc, and ML Engineer agents within a simulated lab structure. [The Virtual Lab](#) uses an LLM Principal Investigator to orchestrate specialized scientist agents, producing 92 novel nanobody binder designs for SARS-CoV-2. These systems are part of an early-stage trend toward multiagent coordination in biomedical research, though their outputs still require experimental validation.

